Q4 Quality & Cost Analysis: Detailed Analysis

Supporting Report to the Executive Summary

This report explains what we found in the Q4 quality and cost analysis, how we reviewed the data, and what the findings mean for our business. It also provides the supporting numbers, data concerns, and recommendations to help SHP decide where to focus its efforts in 2026.

The analysis focuses on three main questions:

1. Who are our highest-risk members?
2. Are our care management programs effective?
3. Where should we focus our resources in 2026?

Data review and how I did the analysis:

I used Pivot Table in Excel to verify the count of records and members.

File	Rows	Members	Used for
claims.csv	5,100	994	Plan-paid cost (paid_amount), service category, diagnosis, denials, providers
encounters.csv	3,000	946	ED visits, inpatient admissions, readmission flag, preventive visits, chronic-condition flags
care_mgmt.csv	400	243	Enrollment, engagement, program outcomes
um.csv	800	530	Prior authorization decisions, turnaround, appeals
ops.csv	100	n/a	10 operational metrics x 10 months; reconciled to detail data

Key assumptions

- **The extract is a ~1,000-member sample.** Member IDs run 1 to 1,000 against ~250,000 covered lives. I report relative patterns and per-member rates and never scale dollar totals to the full book.
- **Window is 1 Jan to 15 Oct 2025 (288 days, ~9.5 months).** No eligibility file exists, so every member is assumed continuously enrolled for the window. Rates per member-month use 9.5 months.
- **Cost = plan-paid amount** (paid_amount). Denied claims are $0 paid. PENDING claims already carry paid amounts and are included.
- **Inpatient utilization** is counted from INPATIENT_ADMIT encounters only (discharge rows would double count).
- **Chronic conditions** are the union of encounter chronic_conditions flags and chronic diagnosis codes on claims (diabetes E10-E11, cardiac I20-I25/I48/I50, hypertension I10-I13, asthma J45, COPD J44).
- **Care-management enrollment** ("ever" or "active") uses the care_mgmt file as given. Members can be in several programs; member-level analysis uses the first enrollment.

Data quality issues found

During the review I identified several issues that could affect the results

Area	What We Found	How We Handled It
Member IDs	Encounter IDs had an extra two-digit suffix.	Removed the extra digits to match members across files.

Membership	Only about 1,000 members were included, not the full 250,000.	Treated the data as a sample and did not estimate full-population costs.
Eligibility	No enrollment dates or member-month information was available.	Assumed continuous enrollment for 9.5 months.
Claims timing	Claims activity ended October 15, but payments continued through November 28.	Used service dates for trends and flagged recent months as incomplete.
Denied claims	Denied claims showed member responsibility amounts even though the plan paid $0.	Excluded member responsibility from plan-paid costs.
Pending claims	Some pending claims already had payments recorded.	Included them as reported. Excluding them would reduce total costs by 2.6%.
Diagnosis coding	Some hospital claims had diagnoses that did not appear to match an inpatient stay.	Kept the original data but flagged it for clinical review.
Hospital records	There were more inpatient claims than admission encounters.	Used encounters for admission counts and claims for cost analysis.
Readmissions	The readmission flags did not match what could be calculated from the available hospital records.	Used the reported flags but recommended checking the calculation.
Care-management records	Some enrollment dates, completion statuses, and risk scores were inconsistent.	Applied consistent rules and documented the differences.
Repeat enrollment	Some members had multiple records for the same program, including overlapping dates.	Kept the records but flagged them for cleanup.
Operational reporting	Some monthly performance labels did not match the actual results.	Recalculated performance using the correct direction for each metric.

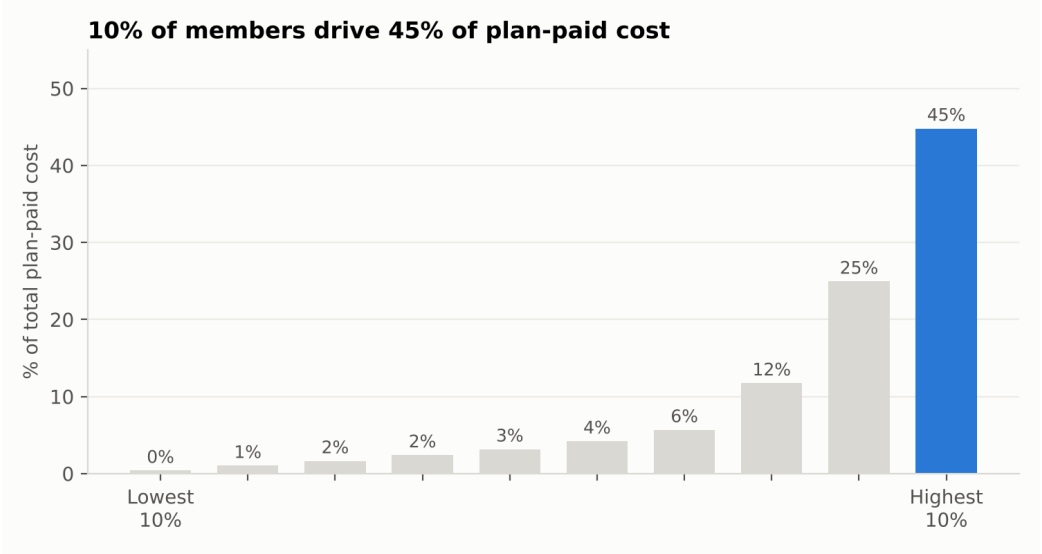
1. Who are the highest-risk members?

1.1 A Small Group of Members Drives Most of the Cost

One of the clearest findings is that healthcare spending is concentrated among a relatively small number of members.

Total plan-paid claims were approximately **$9.12** million for the sample.

The average cost per member was $9,119, while the median was only $3,275. This tells us that a smaller number of members had very expensive healthcare events that increased the overall average.



What this tells us: The top 20% of members account for approximately 70% of healthcare spending. This is where targeted outreach and better care coordination could have the greatest impact.

1.2 How I Identified High-Risk Members

Since there was no existing definition of high risk, we created a simple scoring method using four areas:

1. Healthcare spending
2. Hospital admissions
3. Emergency room visits
4. Number of chronic conditions

Members received points based on their activity. Members with more points were placed into higher-risk groups.

Risk Category	Number of Members	% of Members
Low	646	64.60%
Moderate	281	28.10%
High	61	6.10%
Very High	12	1.20%

The High and Very High groups represent only 7.3% of members but account for approximately 26% of total healthcare spending.

These members also have significantly more hospital admissions than members in the Low-risk group.

An important finding is that 37 of the 73 High and Very High members have never been enrolled in care management.

This gives SHP an opportunity to improve how we identify and reach members who may benefit from additional support.

The Moderate group is also important. It represents 28% of members and 39% of total costs. These members may benefit from less intensive programs before their healthcare needs become more complicated.

1.3 Understanding the Different Member Groups

We also divided members into six groups based on their healthcare needs and utilization.

Segment	Members	% of cost	Paid / member	Admits /100	ED /100	Avg chronic	Ever in CM
A. Hospital-driven high cost	22	9.5%	$39,489	118	32	0.8	41%
B. Frequent ED users	8	0.5%	$6,162	38	338	0.8	25%
C. Multi-condition chronic	184	17.6%	$8,730	21	36	2.5	70%
D. Single chronic condition	164	17.1%	$9,496	20	28	1.0	63%
E. Low-need, engaged in primary care	554	49.4%	$8,131	23	37	0.0	0%
F. Low-need, no primary-care touch	68	5.8%	$7,828	18	32	0.0	0%

The hospital-driven group stands out because only about 41% have ever participated in care management, and only about 23% are currently active.

This suggests that our outreach process may not be reaching some of the members with the greatest need.

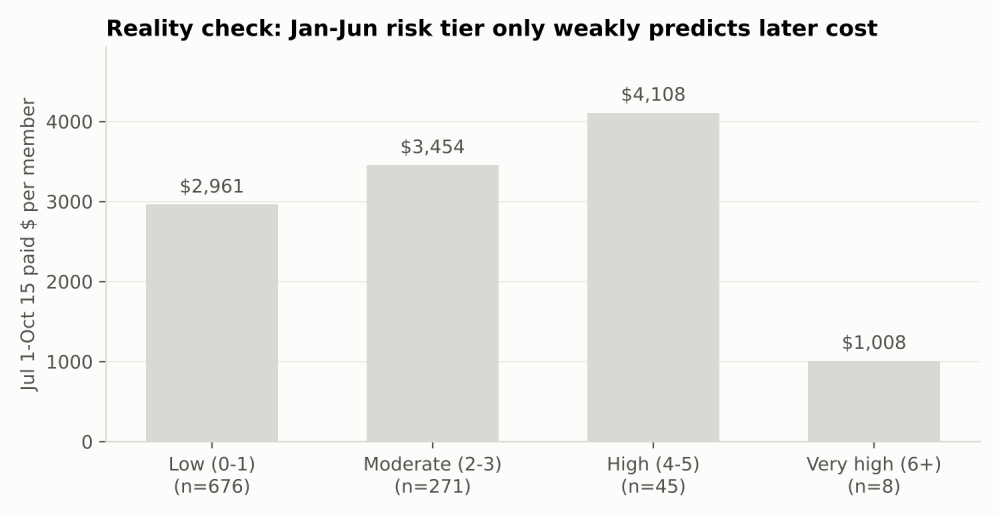
The chronic-condition groups also deserve attention. Although many are already enrolled in care management, their healthcare costs are not dramatically different from those of members without chronic conditions.

This raises an important question: Are our programs helping members manage their conditions before they need more expensive care?

We need better outcome tracking to answer that question.

1.4 Does the risk definition predict the future?

I tested the score prospectively: build it from January-June data only, then compare July 1-October 15 cost. The result is sobering. Member-level correlation between first-half and second-half cost is -0.00; the correlation of score with later cost is +0.02. High-tier members (n=45) cost $4,108 per member later versus $2,961 for Low, a weak 1.4x lift on a small group. In this extract, high cost is largely episodic (single large admissions) rather than persistent.



Implication: the tiers describe where cost and utilization are concentrated now. They are good for triage but should not be sold as a forecasting model. Outreach should rely on clinical triggers (a discharge, repeat ED use, a readmission, uncontrolled chronic disease) that identify members at a moment of need, and any predictive model needs validation on multi-year, full-population data.

1.5 Priority outreach list

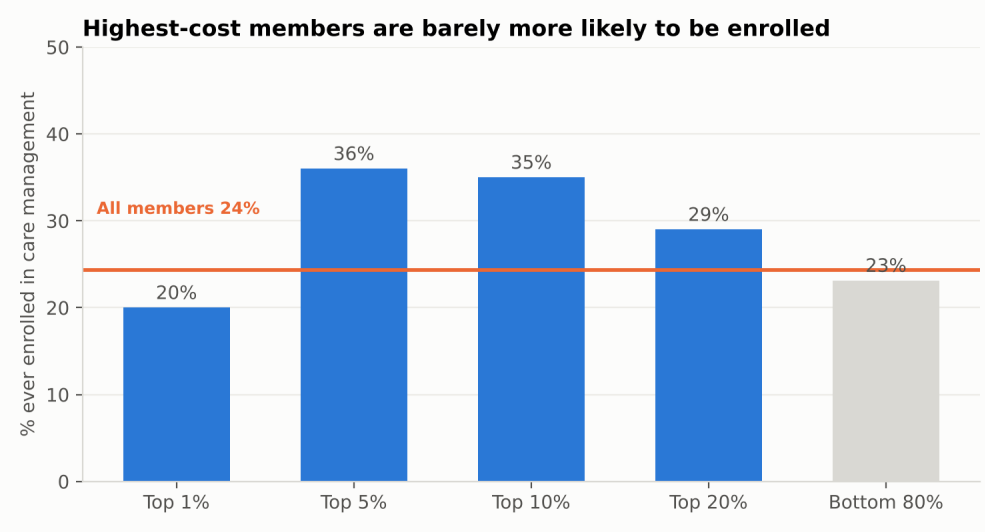
outputs/priority_outreach_list.csv lists the 46 High/Very-high members not currently active in care management, with tier, utilization, conditions and a suggested program (High-Risk Case Management: 20, Cardiac Care: 12, Asthma Control: 7, Diabetes Management: 7). 37 of the 73 High/Very-high members have never been enrolled.

2. Are care-management programs effective?

Effectiveness cannot be read from a simple enrolled-versus-not comparison, because enrolled members are chosen for being sicker. I used four lenses, from weakest to strongest evidence.

2.1 Who is enrolled (targeting)

Member group	Members	% ever enrolled	% active now	Share of cost
Top 1%	10	20%	10%	8%
Top 5%	50	36%	28%	27%
Top 10%	100	35%	27%	45%
Top 20%	200	29%	22%	70%
Bottom 80%	800	23%	18%	30%



Enrollment is only modestly concentrated in high-cost members, consistent with the stated capacity constraints but not with a program aimed at the costliest. The risk score recorded at enrollment is essentially unrelated to later cost (-0.03) or ED use (+0.01). Mean scores by program are Asthma Control 5.4, Cardiac Care 6.3, Diabetes Management 7.1, High-Risk Case Management 5.2. Every enrolled member has chronic-condition evidence, whereas only 16% of non-enrolled members do, so chronic flags in the encounter data appear to be recorded mainly for enrolled members. This limits any like-for-like comparison.

2.2 Cross-sectional comparison

Per member-month (9.5-month window)	Enrolled	Not enrolled	Enrolled vs not, chronic only
Members	243	757	119
Plan-paid $	$1,076	$926	$962
ED visits per 1,000	33.9	41.0	45.2
Admissions per 1,000	27.4	25.0	22.2

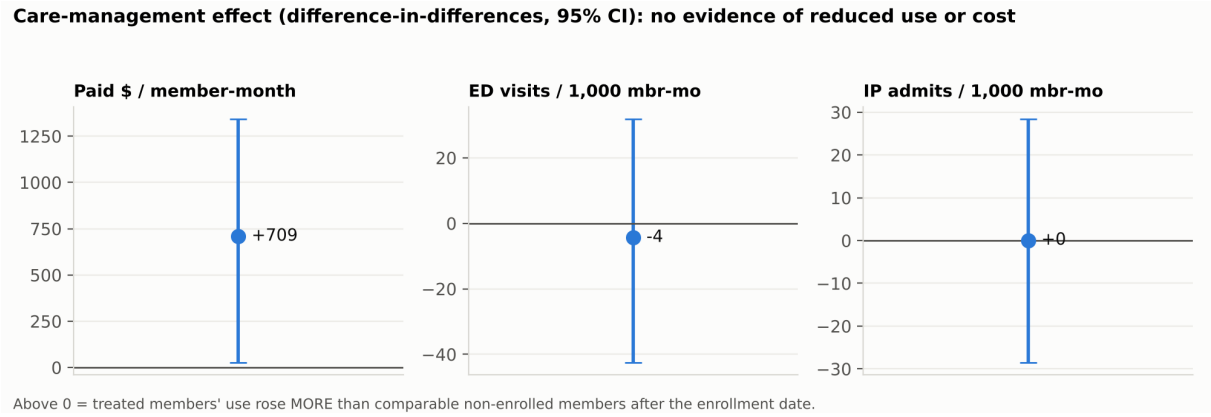
Enrolled members have lower ED rates (rate ratio 0.83, 95% CI 0.64-1.06 versus all non-enrolled; 0.75, CI 0.53-1.07 versus non-enrolled members with chronic conditions) but the intervals include 1, and cost and admissions are similar or slightly higher. The third column shows the more comparable group (members with chronic conditions); columns two and three together are the 'enrolled versus similar' view. This is directionally encouraging on ED use but not conclusive.

2.3 Before/after versus comparison group (difference-in-differences)

Design. Treated: 68 members whose first-ever enrollment fell between 1 Feb and 31 Jul 2025, so each has at least one month of claims before and ~2.5 months after. Comparison: the 757 never-enrolled members, each assigned a random pseudo enrollment date drawn from the treated dates (200 draws averaged; 600 bootstrap resamples for intervals). The outcome is the change in monthly rate from before to after, treated minus comparison. Costs are also shown with claims capped at the 99th percentile ($31,694) so single claims cannot dominate.

Outcome	Treated before	Treated after	Comparison before	Comparison after	Difference-in-differences	95% CI
Paid $ per member-month (capped)	$563	$1,153	$973	$854	+709	27 to 1,343
Paid $ per	$586	$1,223	$1,002	$879	+761	72 to 1,495

member-month (uncapped)						
ED visits per 1,000 member-months	38.3	39.5	37.3	42.8	-4.3	-42 to 32
Inpatient admits per 1,000 member-months	19.8	22.2	23.3	25.6	+0.1	-28 to 28



Reading it. ED visits and admissions show no detectable change. Spending rose more for treated members (+709 per member-month, CI 27 to 1,343), but I do not treat this as evidence that programs raise cost: (a) the treated group had low pre-period spending ($563 versus $973 for comparators) so some rebound is expected; (b) removing the five treated members with the largest increases cuts the estimate to +178; (c) four outcomes were tested and the cost interval barely excludes zero; (d) enrollment is triggered by clinical events, and care managers may uncover needed care. Members with an admission rose from 6% before to 10% after, which is only 4 vs 7 people.

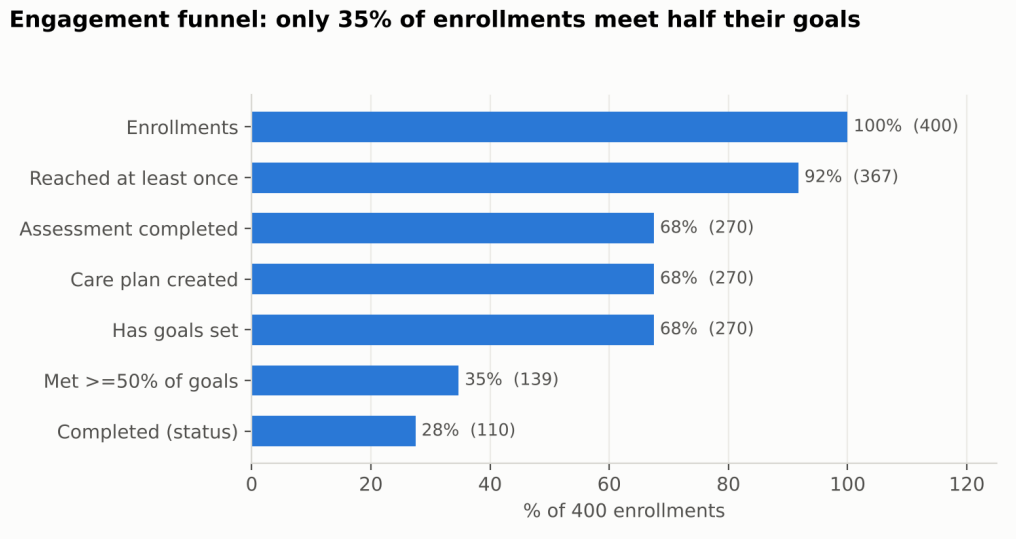
Bottom line: no evidence of savings, no reliable evidence of harm, and not enough data to tell. A sample of 68 members with a short, lumpy pre-period cannot detect an effect of realistic size.

2.4 Does more engagement mean better results?

Engagement marker	Group	n	Change in $ per member-month	Change in ED per 1,000 member-months
Care plan created	Yes	49	$410 (SE $335)	-4.5
Care plan created	No	19	$1,055 (SE $771)	+15.7
Contact rate >= 50%	Yes	37	$796 (SE $317)	-5.7
Contact rate >= 50%	No	31	$345 (SE $601)	+9.3
Goal attainment >= 50%	Yes	23	$308 (SE $492)	-1.6
Goal attainment >= 50%	No	45	$735 (SE $419)	+2.6

Members with a care plan and assessment (these two always coincide in the data) saw ED visits fall slightly (-4.5 per 1,000) versus a rise (+15.7) for those without, and smaller cost growth, but the standard errors are as large as the differences. Directionally consistent with engagement helping; statistically inconclusive.

2.5 Program operations

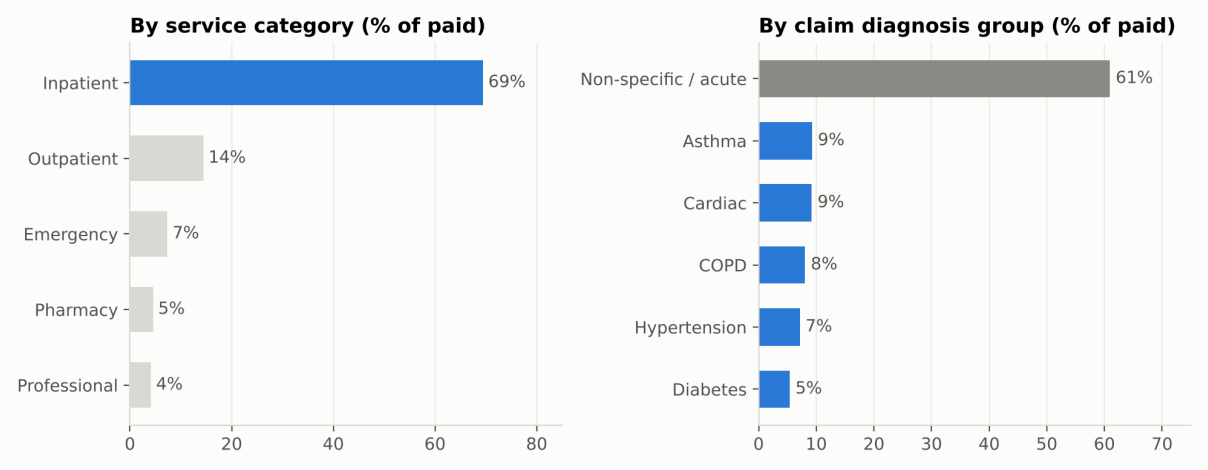


Program	Enrollments	Members	Avg risk score	Contact rate	Care plan	Goal attainment	% completed	% disenrolled	% active
Asthma Control	87	49	5.4	47%	66%	44%	29%	15%	56%
Cardiac Care	121	69	6.3	47%	66%	46%	27%	12%	60%
Diabetes Management	115	78	7.1	50%	68%	48%	30%	10%	60%
High-Risk Case Management	77	47	5.2	51%	71%	44%	23%	8%	69%

- **Contact rate is 49% of attempts** and overall goal attainment 46%. Differences between programs are small; Diabetes Management has the highest risk scores and goal attainment.
- **Care managers vary widely** (goal attainment 27%-62%, contact rate 33%-55%) but with only 10-22 enrollments each, this is partly chance. Worth monitoring with more data.
- **Stale enrollments.** Median tenure of an active enrollment is 311 days; 81 are over a year old and 187 over 180 days. 65 of the latter have no care plan or completed assessment, so they consume caseload without activity.
- **Re-enrollment.** 157 enrollments repeat a member-program pair, and 128 of them overlap in time with a previous enrollment in the same program. That looks like duplicate records and should be checked.
- **Closure.** Of closed enrollments, 71% are COMPLETED, but program_completion_flag is TRUE on only 44 records, so the two fields disagree.

3. Where Should SHP Focus in 2026?

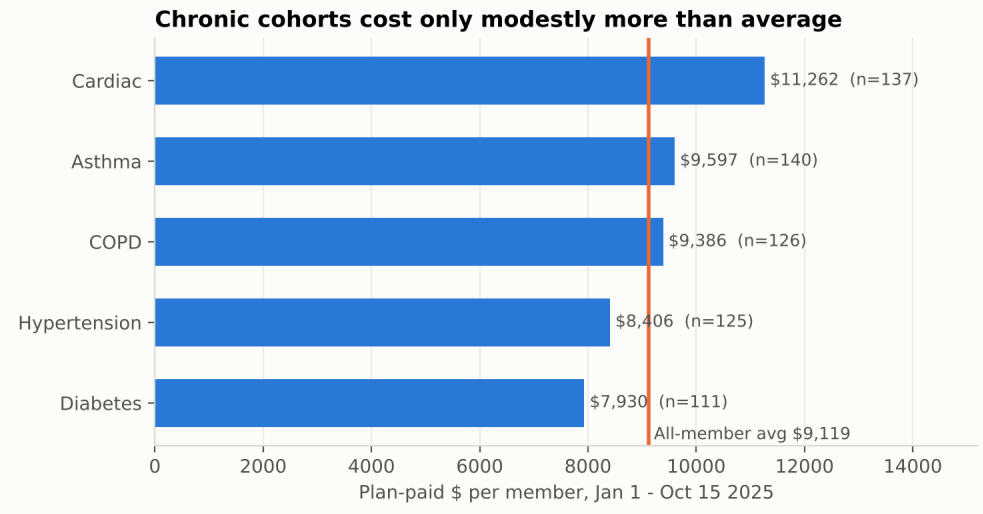
3.1 What drives cost



Inpatient care is 69% of cost from 310 claims averaging $20,436. Emergency (7%) and outpatient (14%) follow; pharmacy and professional services together are under 9%. Claims coded to the five target chronic conditions account for 39% of cost; 61% is coded to non-specific or acute diagnoses (fever, abdominal pain, headache, URI, GERD, back pain, routine exam).

3.2 Chronic conditions

Condition cohort	Members	Paid / member	vs other members	% of total cost	Admits /100	ED /100	Readmits	In CM	Preventive visit
Cardiac	137	$11,262	1.28x	16.9%	18	34	2	61%	35%
Asthma	140	$9,597	1.06x	14.7%	27	34	6	66%	39%
COPD	126	$9,386	1.03x	13.0%	25	36	4	71%	40%
Hypertension	125	$8,406	0.91x	11.5%	24	42	2	68%	34%
Diabetes	111	$7,930	0.86x	9.7%	30	34	5	70%	33%



Cohorts overlap, so shares sum above 100%. **Cardiac** is the largest driver ($11,262 per member, 1.28x others, 16.9% of cost), followed by asthma and COPD near the average. Diabetes and hypertension cohorts cost less than other members in this window. Members with at least one chronic condition (362) hold 39% of cost,

only slightly more than their 36% share of members, so chronic disease alone is not what makes a member expensive here. Preventive-visit rates among chronic members (38%) are not higher than among others (35%), a potential quality gap for the care-management programs.

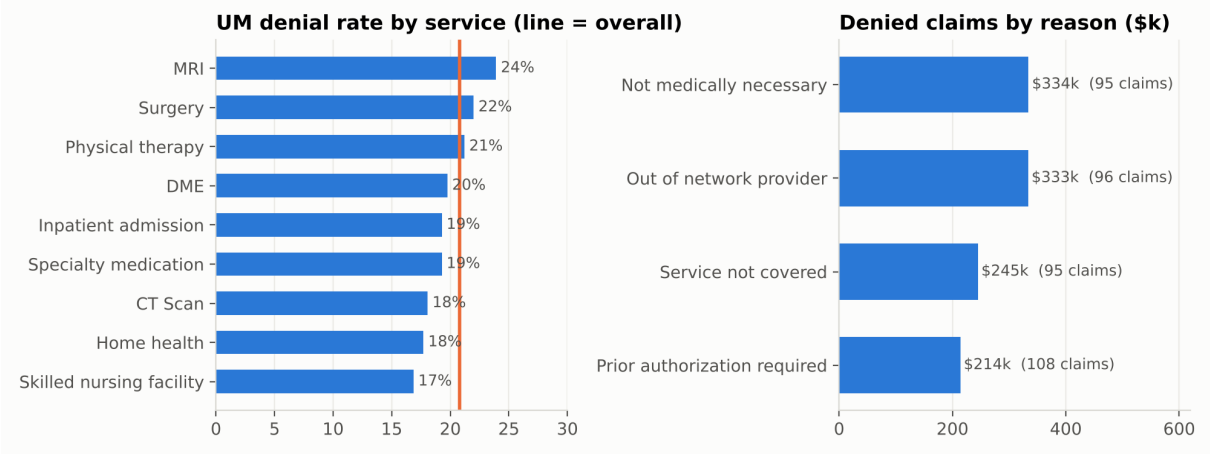
3.3 Inpatient diagnosis review

62% of inpatient claims ($3,752,609) carry a non-specific symptom or routine diagnosis. 28 are coded Z00.00 (routine exam, $488,609), and with URI (J06.9) and headache (R51) the pool is **83 claims and $1,604,677 (18% of all paid cost)**. These diagnoses rarely justify an inpatient stay. The cause could be coding of the principal diagnosis, admissions that did not need an inpatient level of care, or an artefact of the data. A sample clinical-validation audit would settle which. If even a fraction is avoidable or mis-billed, this is the biggest savings opportunity found.

3.4 Providers and facilities

The top 20 of 200 providers account for 24% of paid cost and the top 10 of 50 facilities for 36% of inpatient cost. To see whether any provider is a real outlier I compared each provider's paid cost with the average for the service categories they billed. Observed spread in these ratios (SD 0.310) is no larger than chance reshuffling of claims within category (SD 0.287; 95th percentile 0.305). Only 1 provider has a denial rate significantly above the 7.7% average at $p<0.01$, about what chance gives across 200 providers. Facilities have a median of 5 admissions each, too few to rank on readmissions or length of stay. **The data do not support naming specific providers or facilities**; provider review should wait for the full-population data, using this method.

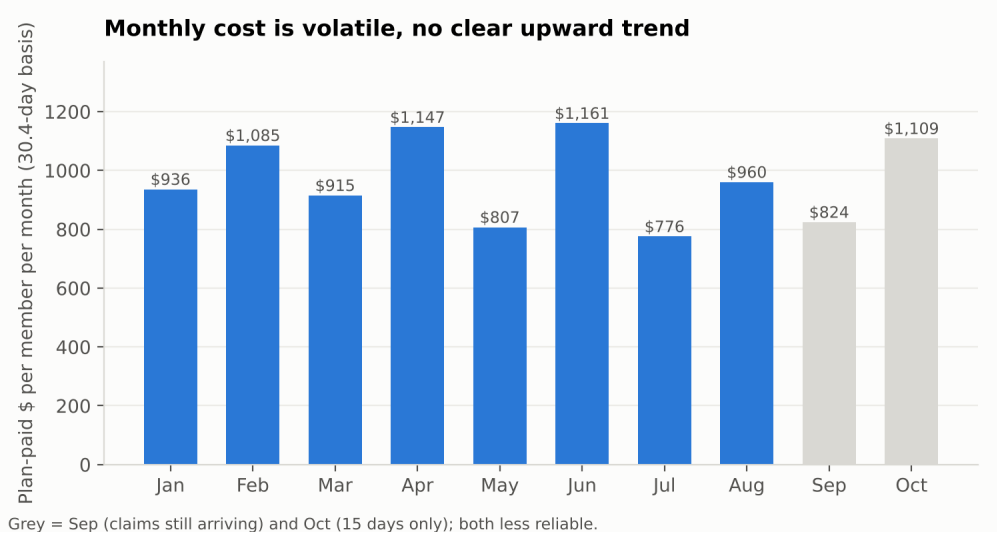
3.5 Claim denials and utilization management



- **Claim denials:** 394 claims (7.7%) denied, $1,126,274 allowed value. Reasons are spread evenly; missing prior authorization is the most frequent (108 claims) and out-of-network the second largest by value ($333,441). These are process failures that provider education and network steerage can address.
- **UM decisions:** 21% of decided requests are denied (MRI 24%, surgery 22%). 47 of 159 denials were appealed (30%) and **41% of decided appeals were overturned**, which suggests some initial denials are not holding up.
- **Turnaround:** average 4.0 days; urgent and emergency requests average 3.8 days and 53% take longer than 3 days. Urgent pre-service decisions commonly carry a 72-hour standard; SHP should check its applicable state and federal requirements.

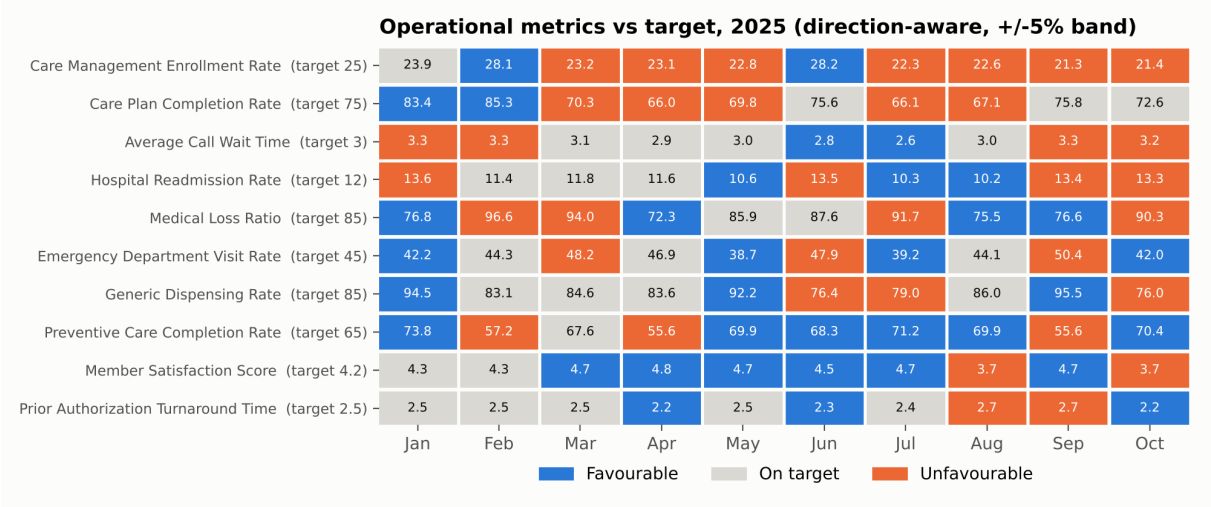
3.6 ED use, preventive care and cost trend

303 members had an ED visit, but only 8 had three or more (7% of visits), so repeat-ED outreach is a very small target here. Only 36% of members had a preventive visit in the window. Monthly cost is volatile (chart below) with no clear upward trend; January-June averaged $998 per member-month and July-September $861, though recent months are incomplete because claims arrive about 27 days after service. The scenario states costs rose above budget; no budget or premium data was provided, so that claim could not be tested.



Grey = Sep (claims still arriving) and Oct (15 days only); both less reliable.

4. Operational metrics and reconciliation



Re-derived with explicit direction and a ±5% band: care-management enrollment rate is unfavourable in 7 of 10 months and care-plan completion in 5; call wait time, readmission rate and medical loss ratio each in 4. Average MLR is 84.7% against an 85% target, exceeding it in 6 of 10 months. The ops file labels MLR as if higher were better (a 96.6% MLR is marked ABOVE_TARGET); I treated lower as favourable.

Reported metrics were recomputed from the detail files where possible:

Metric	Recomputed	Reported in ops.csv	Comment
ED visits per 1,000 per month	37.2	44.4	Recomputed is ~16% below ops (ops may cover a wider population)
Members active in CM, % (monthly	20.2	23.7	Ops ~23.7% matches 24.3% ever-enrolled, not the

avg)			20.2% active in a month
Readmission rate %	8.0	12.0	Ops 12.0% vs 8.0% monthly (7.4% overall); flag cannot be reproduced
Prior-auth turnaround, days	4.0	2.5	Ops understates by ~1.5 days (calendar vs business days?)
Preventive care, %	36.2	66.0	36% of members have a preventive encounter vs 66% reported; definitions likely differ
CM enrollment, % ever enrolled	24.3	23.7	Consistent with ops (supports the ~1,000-member sample assumption)

5. Limitations and next steps

- No member eligibility, demographics, premiums or budget: no member-months, PMPM, MLR or age/sex adjustment, and the 'cost above budget' premise cannot be tested.
- Claims start in 2025 but half of enrollments began in 2024: no pre-enrollment baseline for most members, which is why effectiveness rests on 68 members.
- Chronic flags seem to be recorded mainly for care-management members, which biases comparisons.
- Utilization in the extract is far above typical commercial levels (about 307 admissions and 471 ED visits per 1,000 per year; roughly 5x and 2-3x common benchmarks, to be confirmed with SHP's own), so it may over-represent heavy users.
- Diagnosis codes look weakly related to setting, so condition-level and inpatient-diagnosis findings need clinical validation.
- Next steps: request the eligibility file and 2024 claims; rerun the evaluation with a 12-month matched design; run a sample audit of low-acuity inpatient admissions; reconcile the ops definitions with the data team.

Questions for the stakeholder: Is the extract random or enriched for utilizers? How does ops compute preventive-care completion and prior-auth turnaround? Is there an enrollment-capacity target per program? Can we get member eligibility data?